

Discussion of Wealth Inequality and Safe Asset Demand

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A Macro Reading

- **Excellent paper**: very instructive, conceptually clear, a joy to read!
- The language is asset pricing; the core issue is **heterogeneous-agent macro**.
- The broad force is familiar: idiosyncratic risk raises safe-asset demand.
- Main (technical) question: when does individual optimality aggregate?

individual TVCs hold $\nRightarrow$ aggregate TVC holds

- My macro reading: the paper sharpens an old question
 - When do **Bewley-Huggett-Aiyagari (BHA)**-style forces break $P = \text{PV}(\text{cash flow})$?
 - The answer lives in the **left tail of marginal utility**.

$\Rightarrow$ **An aggregation theorem, a bubble result, and a diagnostic for HA models.**

Government Debt: A Useful Decomposition

Focus on government debt for exposition

- Let b_t be the market value of debt and s_t the primary surplus.
- Iterating the government budget constraint gives:

$$\underbrace{b_t}_{\text{market value of debt}} = \mathbb{E}_t \underbrace{\int_t^\infty M_{t,t+s}^i s_{t+s} ds}_{\text{PV of future primary surpluses}} + \underbrace{\lim_{T \rightarrow \infty} \mathbb{E}_t [M_{t,T}^i b_T]}_{\text{rational bubble term}}.$$

- The last term is the **rational bubble term = issuer's (buy-and-hold) TVC**.
- Each *household's own* TVC always holds (optimality); the failure is at the **aggregate** level.
- If the term vanishes, debt backed by future surpluses. Otherwise, rollover bubble.

Why Bounded Risk Rules Out the Bubble

Individual TVCs aggregate when nobody can fall arbitrarily far behind

- The SDF depends on **consumption growth**.
- For aggregate debt, the relevant risk is **relative** consumption growth:
 - can household i become tiny relative to the economy?
- $\eta_T^i \equiv$ household i 's consumption growth relative to agg. cons. growth from 0 to T .

$$\eta_T^i \geq \varepsilon > 0 \text{ for all } i, T \implies \lim_{T \rightarrow \infty} \mathbb{E}_0[M_{0,T}^i b_T] = 0.$$

- Standard **BHA**: y^i, c^i bounded $> 0 \Rightarrow$ **aggregate TVC holds**.

$\implies$ **With a consumption floor, the literal asymptotic bubble mechanism disappears.**

Where This Fits: BHA vs. DJ

Same family of mechanism; different limit

- **Bewley-Huggett-Aiyagari (BHA)** tradition has long delivered:
 - shape of wealth distribution $\rightarrow$ safe-asset demand
 - idiosyncratic risk depresses market r^f below ρ ; **even $r^f < g$ possible**
 - does **not require a rational bubble**

Bewley '86, Huggett '93, Aiyagari '94, Aiyagari-McGrattan '98, Krueger-Lustig '10, Hubmer-Krusell-Smith '21, Reis '21, Kocherlakota '23, Aguiar-Amador-Arellano '24.

- **DJ** sharpen on a related but distinct question:
 - drop the consumption floor: $\mathbb{E}_0[(\eta_T^i)^{-\gamma}]$ can outrun $e^{\rho T}$
 - **asymptotic** aggregate-TVC failure $\Rightarrow$ **infinite bubble term**
 - the buy-and-hold present-value formula itself fails

$\Rightarrow$ **Same precautionary force; BHA a finite premium, DJ an asymptotic limit.**

When Can the Bubble Survive?

For exposition: log utility, no aggregate risk, $g = 0$

■ Two rates:

1 market r^f = one-period equilibrium safe rate

2 r^{agg} = long-horizon decay rate of $\underbrace{\mathbb{E}_0[m_T^i \bar{n}_T]}_{\text{PV of period-}T \text{ agg. wealth}}$. Aggregate TVC holds iff $r^{\text{agg}} > 0$.

■ **DJ:** $r^f = r^{\text{agg}} = \rho - \tilde{\sigma}^2$ (geometric Brownian motion ties them together)

bubble survives iff $\underbrace{\rho}_{\text{effective discount rate}} < \underbrace{\tilde{\sigma}^2}_{\text{variance of idiosyncratic permanent income shocks}}$

■ **BHA:** $r^f < r^{\text{agg}} = \rho$ (bounded c pins r^{agg} at ρ)

⇒ Both can deliver $r^f < 0$, same deficit arithmetic; different valuation logic.

– **BHA:** $\underbrace{b_t}_{> 0, \text{ finite}} > \underbrace{PV_t(s)}_{< 0, \text{ finite}} + \underbrace{TVC_t}_{= 0}$ (finite liquidity wedge)

– **DJ:** $\underbrace{b_t}_{> 0, \text{ finite}} = \underbrace{PV_t(s)}_{= -\infty} + \underbrace{TVC_t}_{= +\infty}$ (terminal term survives)

What Does the Bubble Term Mean?

■ Government debt

- Cash flow in the issuer valuation is the primary surplus s_t .
- In the strict DJ case, the bubble term is $+\infty$.
- Finite debt value requires PV of fiscal backing $= -\infty$: persistent deficits rolled forever.

■ Positive-dividend aggregate claims

- No negative cash-flow PV to offset an infinite bubble term.
- Asset can become infinitely valuable: standard no-trade equilibrium may fail to exist.
- Constantinides-Duffie warning.

Agg. TVC fails $\implies$ rational bubble for debt, non-existence for positive-dividend claim.

Macro Translations: Wealth and Risk

What “wealth” and “transitory” mean here

- **Wealth.** In **DJ**, n^i is a consumption-relevant resource state ($c^i \propto n^i$).
 - In standard **BHA**, wealth typically means *marketable net worth*: excludes human wealth.
 - Low measured net worth $\nrightarrow$ near-zero consumption.
 - Empirical question: does the wealth left tail map to the consumption left tail?
- **Risk.** In **DJ**, “transitory” = permanent multiplicative risk + Poisson reset/death.
 - Cross-sectional distribution stationary because of Poisson reset/death (perpetual youth).
 - In standard **BHA**, “transitory” = mean reversion in labor income (e.g. AR(1)).
 - Stationary-via-reset $\neq$ ordinary transitory income risk.

For empirical relevance: measure consumption growth left tail, not wealth left tail.

What Would Discipline the Mechanism?

- With CRRA, consumption floors (social insurance) rule out literal asymptotic bubble.
- More generally, the condition is about **marginal utility** exploding.
 - Even if $c_t^i > \varepsilon > 0$, can have $u'(c_t^i) \rightarrow \infty$ (e.g. subsistence as in Stone-Geary, habits)
- Implication I: **high shadow returns** to liquidity in the left tail (e.g. payday loans).
- Implication II: households exposed to these states are on **ex-ante** asset-demand margin.
 - Example: “*AI Doom*” scenario (today’s median hh expecting very low future c)

Current bondholders need not be poor ex post; they must face the left-tail risk ex ante.

Take-Aways

For heterogeneous-agent asset pricing

- **Conceptual:** **BHA** precautionary force pushed to its asymptotic limit.
 - When $c \rightarrow 0$ is in the support, the buy-and-hold PV formula can fail.
 - **Theoretical:** Diagnostic / warning for HA exercises that impose TVC.
 - Constantinides–Duffie and broader HA asset-pricing literature.
 - **Empirical:** Mechanism is literal only when marginal utility $u'(\cdot)$ can explode.
 - With CRRA: this means $c \rightarrow 0$.
 - With subsistence / habits: the effective consumption gap can go to zero.
 - Right object: lower-tail consumption / $u'(\cdot)$ growth, not wealth Gini.
- ⇒ **Left-tail risk can shift the *valuation regime*, not just lower r^f .**